Conditional Generation of Lithium-Vacancy Configurations for Cathode Relithiation

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1 Research problem

Lithium cobalt oxide ($LiCoO_2$, LCO) is a battery cathode (positive electrode) material; vacancies are empty lithium sites. Many arrangements share the same lithium content, making exhaustive energy evaluation impractical. We ask: **Can a compact conditional variational autoencoder (CVAE) discover new, diverse, low-energy vacancy configurations more efficiently than simple sampling methods under the same evaluation budget?** We will generate binary site occupations and reconstruct three-dimensional structures in one fixed crystal framework.

2 Scientific importance

Hydrothermal relithiation replenishes lithium using hot, pressurized, lithium-containing water [1]. Our main study aims to map reactor temperature, pressure, solution composition, and initial cathode lithium content to *thermodynamic endpoints*: the final lithium contents favored at equilibrium. Pairing these predictions with experimental relithiation rates would support computational recommendations for optimal conditions not yet tested experimentally. Lithium/vacancy arrangements have different energies [2]; identifying representative configurations at each lithium fraction is therefore a necessary input to this reactor study.

3 Existing work and limitations

Park et al.'s ChargeDIFF combines a vector-quantized variational autoencoder, which compresses the spatial distribution of electrons, with diffusion that jointly generates crystals and their electron densities [3]. Guiding generation toward channels with low electron density produced battery-cathode candidates with targeted lithium-migration pathways; the authors then examined lithium motion after introducing a vacancy. This motivates our focus on generating vacancy arrangements within one LCO framework.

To evaluate such structures, reference data must connect atomic geometry to physical properties. Barroso-Luque et al.'s Open Materials 2024 (OMat24) pairs 118 million atomic structures with energies, atomic forces, and cell stresses calculated using density functional theory (DFT), a quantum-mechanical method [4]. Its equilibrium and distorted structures link atomic positions and cell geometry to energetic and mechanical responses.

These paired data train machine-learning interatomic potentials (MLIPs) to approximate that relationship. Wood et al.'s Universal Models for Atoms (UMA), trained on OMat24 and other DFT datasets, predicts energies, forces, and stresses from atomic identities, positions, and the cell [5]. Evaluating new structures then requires no new DFT calculation, making repeated screening of generated vacancy configurations practical.

ECE285: Deep Generative Models, UC San Diego. Fall 2026 project proposal.

4 Proposed solution and evaluation

Data and representation. We will select up to three lithium fractions from 24,145 LCO configurations with existing UMA energy predictions. Binary masks encode 144 available lithium sites: 1 for vacant and 0 for occupied. The occupied-site count divided by 144 gives the lithium fraction; the cell and cobalt/oxygen coordinates remain fixed. UMA energies define binary low-energy labels from the bottom 20% of *training* energies within each fraction, connecting structure generation to energy-based selection.

Model and training. The CVAE conditions on lithium fraction and energy label, using two small fully connected layers, a Gaussian latent dimension of 4–16 selected on validation data, binary reconstruction loss, and Kullback–Leibler (KL) regularization toward a standard Gaussian. Adam training uses validation evidence lower bound (ELBO) for early stopping. Initial KL annealing gradually increases regularization to reduce posterior collapse (ignoring latent variables). An unconditional VAE trained on the same data tests conditioning. CVAE generation requests the low-energy class; both VAEs select the required number of highest-scoring sites as vacancies. We report validity before/after projection, reconstruct atomic structures, and evaluate them with frozen UMA.

Benchmark. We will split data 70/15/15% for training/validation/testing, grouping symmetry-equivalent patterns and generation families to avoid leakage. We will compare both VAEs with uniform exact-count sampling, low-energy site-frequency sampling without replacement, and vacancy/occupied-site swaps from low-energy training examples. All methods share applicable training information. At 100 attempts per method, three fractions, and three seeds, we obtain at most 4,500 proposals; duplicates count toward the budget. The limit is 5,000 UMA calls including checks. We will measure unique low-energy discoveries absent from the *entire* dataset per attempt, accounting for symmetry, alongside energy distributions, minimum differing-site count from training patterns, vacancy-neighbor/layer statistics, runtime, and seed variability. These outcomes establish sampling utility under UMA; physical accuracy requires independent DFT validation.

5 Novelty, significance, and scope

Our contribution is a budget-matched test of conditional generation and a reusable sampler. Low-energy candidates prioritize DFT calculations; diverse configurations and energies can underpin a *lattice model*, an energy predictor for occupied/empty sites. Equilibrium sampling could add *configurational entropy*—the thermodynamic contribution from multiple arrangements—to the main study's data [6]. Higher-energy arrangements also contribute; CVAE frequencies are not equilibrium probabilities. Optional DFT labeling under equal generated/baseline budgets could support MLIP fine-tuning, tested on independent DFT data. These extensions remain outside the core class commitment.

6 Tentative timeline and responsibilities

We budget 2–3 hours per person weekly, including meetings and writing. Ricardo (R) owns chemistry, data export, scoring, and physical interpretation; Beihao (B) leads the CVAE; Varad (V) leads baselines/evaluation. All share theory, reporting, and presentation.

Dates (2026)	**Milestone and lead responsibilities**
Oct 19–25	Proposal due Oct 21, 11:59 p.m. PT; prepare mask/energy examples (R), CVAE prototype (B), and uniform sampler (V).
Oct 26–Nov 1	Validate energies (R), add conditioning (B), group data and build baselines (V).
Nov 2–8	Tune latent size (B), test metrics (V), run three seeds (all); freeze settings.
Nov 9–15	Score up to 4,500 candidates (R); compare VAEs (B) and sampling baselines (V).
Nov 16–22	Select DFT candidates (R), write methods (B), plot comparisons (V); freeze results.
Nov 23–29	Draft report, including reactor modeling implications (all); two hours each.
Nov 30–Dec 4	Finalize code/report/slides and rehearse (all); submit by the instructor's schedule.

References

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[3] Junkil Park, Junyoung Choi, and Yousung Jung. Generative modelling of inorganic materials with explicit electronic structure. *Nature Communications*, 17:7070, 2026. doi: 10.1038/s41467-026-73985-2.

[4] Luis Barroso-Luque, Muhammed Shuaibi, Xiang Fu, Brandon M. Wood, Misko Dzamba, Meng Gao, Ammar Rizvi, Matt Uyttendaele, C. Lawrence Zitnick, and Zachary W. Ulissi. Open materials 2024 (OMat24) inorganic materials dataset and models. *arXiv preprint arXiv:2410.12771*, 2024. URL `https://arxiv.org/abs/2410.12771`. Version 2, revised 2026.

[5] Brandon M. Wood, Misko Dzamba, Xiang Fu, Meng Gao, Muhammed Shuaibi, Luis Barroso-Luque, Kareem Abdelmaqsoud, Vahe Gharakhanyan, John R. Kitchin, Daniel S. Levine, Kyle Michel, Anuroop Sriram, Taco Cohen, Abhishek Das, Ammar Rizvi, Sushree Jagriti Sahoo, Zachary W. Ulissi, and C. Lawrence Zitnick. UMA: A family of universal models for atoms. *arXiv preprint arXiv:2506.23971*, 2025. URL `https://arxiv.org/abs/2506.23971`. Version 2, revised 2026.

[6] M. Wagemaker, A. Van der Ven, D. Morgan, G. Ceder, F. M. Mulder, and G. J. Kearley. Thermodynamics of spinel Li_xTiO_2 from first principles. *Chemical Physics*, 317:130–136, 2005. doi: 10.1016/j.chemphys.2005.05.011.